

INT422 IT Project Management

Unit I

Software Cost and Schedule Estimation

Lecture 12

Outline

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 - Size
 - Productivity
 - Parametric Models
 - Risks
- Scheduling
- Costing
- Putting the Estimate Together
- “Good Ideas” for Improving Estimates
- Tracking Execution
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- Conclusion

Introduction

- The main purpose of the paper is to present approaches for deriving an estimate of the cost and schedule of a software project
- Discusses methods to track and alter the estimates as development progresses
- Discusses ways to get a project back on track after changes have been made to a schedule

Creating an Estimate...

- Estimates
 - Generally focus on labor hours, quantity of materials and amount of services, not the cost
 - This is computed later
 - Requires determining the work required to meet requirements and the effort required to perform that work

Creating an Estimate...

- Step 1: Identify the tasks
 - They fall under four main categories:
 1. Engineering
 2. Program Management
 3. Configuration Management
 4. Quality Assurance
 - Tasks are recorded in a Work Breakdown Structure (WBS)
 - Hierarchically identifies all tasks in a project
 - Each successive layer should be more descriptive than its parent
 - For a software project, the lowest level should be detailed enough to show class names
 - This is not always possible, or even necessary

Creating an Estimate...

- Step 2: Estimate the resources required per task
 - There are many types of resources (that are often billed differently)
 - Materials
 - Subcontracted Items
 - Travel
 - Labor (the biggest one)
 - The focus of the paper is mainly applied to estimating *labor* based on the engineering (development) efforts

Creating an Estimate...

- Step 3a: Estimating the Software Development Effort
 - Basic Method
 - $E = S/P$ (Estimate = Size/Productivity)
 - The hard part is determining the size and productivity variables
 - Estimating Size – three main factors
 1. Units of measure
 2. Software included in the measurement
 3. Amount of reused code
 - Reused code is generally counted differently than newly written code
 - Must track code Added, Changed and Deleted from the reused code

Creating an Estimate...

- Step 3a Continued...
 - Estimating Productivity – An aggregate of the capabilities of the development team
 - Often based on historical project data
 - New project must use the same size measure and must be implemented with equivalent approaches - same programming language, platform, etc.
 - There are a lot of variables that are difficult to quantify that play a role in this estimate
 - Diseconomy of scale – project size and productivity are inversely related

Creating an Estimate...

- Step 3b: Estimating the Software Development Effort
 - Parametric Estimation Methods
 - Some algorithm is used to determine the estimation based on some set of independent inputs
 - Algorithm and Inputs must be created by an expert estimator and tested to fit legacy data
 - Based on theory, experience and expert judgments
 - Algorithms can change between evaluations for the different lifecycle phases or components
 - RA, design, test, etc.

Creating an Estimate...

- Step 3b: Parametric Estimation Methods...Continued
 - Allocations can be automatically made against WBS items to provide schedule detail along with cost
 - Performance:
 - Validation and calibration of the method is very important
 - Models calibrated against general industry data usually provide estimates within 20% of the actuals
 - Models calibrated with an organization's own historical data provide estimates within 5% of the actuals
 - These models **ONLY** provide an estimate of the SW development activities, not the other tasks and items that form a complete estimate

Creating an Estimate...

- Step 4: Estimating Risks
 - Risks are areas that are identified as possible causes of problems in the future
 - Severity is determined by two variables
 - Likelihood of occurrence
 - Impact if it occurs
 - Generally a label of High, Medium or Low is applied to the risk based on those variables
 - Main Risk areas are: Cost, Schedule, Technical and Business
 - During estimation creation a lot of the system risks should become apparent
 - Additional effort should be added to the proposal to track and handle these risks
 - Often taken care of with “Management Reserve”

Scheduling Tasks

- When all of the tasks are identified and decomposed a schedule must be created
 - Generally based on the WBS (if it goes down to the appropriate level)
 - May also be based on outputs of detailed design
- There are often multiple related schedules created with each representing a greater level of detail
 - Highest level shows major milestones
 - A milestone is an event that will occur at a specified date
 - Lowest level shows individual tasks
 - creation of specific classes

Scheduling Tasks

- Dependency checking is important when creating a schedule
 - Some tasks have prerequisites that must complete before they can begin
 - Others are completely independent
 - Which means they can be worked in parallel
- Creating a Schedule does four main things
 1. Sequences tasks
 - Requires analyzing dependencies
 2. Assigns resources to tasks
 - Not specific people, but notional resources
 3. Calculates the length of the tasks
 - Critical Path is the length of time for the longest path through the schedule. This is the program time to complete.
 4. Compares interim milestones with those from the master schedule
- It is important to ensure that the schedule begins and ends cleanly, with no *dangling tasks*

Costing Tasks

- Converts the *effort* calculated previously into actual dollar amounts
 - Must take into account the classification of each person working on the tasks
 - Jr. Engineer, Lead Engineer, Program Manager, etc.
 - Each of these roles are costed at different base amounts
 - So two Jr. Engineers may make different amounts of money, but the customer is charged a single “Jr. Engineer” rate
 - Work is charged based on a *loaded labor rate*
 - This rate (generally per hour) includes not only the cost of the salary for the employee, but additional costs that cover things such as
 - Profit
 - Contracts, IT (and other support departments)
 - Overhead

Putting The Estimate Together

- The final estimate is put together by a business office within the organization
- Inputs are required from lots of others
 - Planners and Engineers define the job
 - Engineers and estimators determine the resources required
 - Business office calculates the real costs
 - Schedulers create the schedule
 - Managers evaluate the results and set the total price
 - They must work in profit and other costs that may affect the project in the future
 - Such as adjustments to labor rates

Improving Estimation Accuracy

- Some “Good Ideas” for improving estimations
 - Understand the requirements
 - Ensure that the appropriate development environment, programming language, etc. are used
 - Collect and use legacy project data
 - Validate the estimation technique against industry or organizational data
 - Mix estimating techniques and see where and why they produce different results

Tracking Expenditures

- Control accounts are created to logically split up the total project funding among the many tasks
- Charge codes are setup so that labor can be charged against the funding in the control accounts
- For overhead and other support purchases there is generally a “buyer” that all requests must go through
 - This allows a greater ability to track expenditures on these types of items

Tracking and Updating

- To track the progress of development three sources of data are used
 1. Overall project plan
 2. Cost accounting data
 3. Project status
- These sources provide inputs into the *Earned Value* variables
 - BCWS – Budgeted Cost of Work Scheduled
 - ACWP – Actual Cost of Work Performed
 - BCWP – Budgeted Cost of Work Performed
 - **ACWP > BCWP = Over Budget**
 - **BCWS > BCWP = Over Schedule**

Managing Estimates During Execution

- Initial estimates are used to acquire initial funding
 - But in software projects these often change throughout the development process
- The progress of the development must be closely tracked to determine when things have gone awry
- When changes must be made the following options are available:
 - Reinterpret the requirements (work with customer)
 - Apply COTS or reuse instead of new build
 - Use automated tools
 - Revise WBS element development resources
 - Change development sequencing
 - Possibly change model to an iterative one
 - Apply additional resources to tasks

Caveats to Using Additional Resources

- Some software components take a minimum amount of time to complete...no matter how many people work on it
 - Insert overused baby in nine months joke here
- Mythical Man Month
 - It is often worse to apply additional resources to a software development team when in a crunch
 - They must be trained
 - They don't have experience with the component
 - Often causes a greater slip in the development
- Additional resources are not free
 - The money to pay for them must come from somewhere, usually another component in the system

CAIV and SAIV

- Cost as an Independent Variable
 - Used to determine what items will be built and when they will be completed based on the funding available
- Schedule as an Independent Variable
 - Used to determine what items will be built and what they will cost based on the schedule that must be met

Conclusion

- **Good primer on estimations of software size and cost**
 - **Left out additional costs such as**
 - **Design**
 - **Test**
 - **Are these “rolled in” with the coding cost in the general case?**
- **End focus on tracking and adjusting cost/schedule was useful, but somewhat out of place**

Questions

Thank you